InfiniRIVER Setup Guide

Overview (v1.0)

The **InfiniRIVER** system is a water framework for unique dynamic river generation based on fluid calculations. The core of the system has an ultra-optimized fluid solver that is applied to a water shader for emulating real time interactions of the water with scene objects.

The system is in Beta and many more features and refinements are planned to be added, most are already included in the asset, like the 3D particles driven by the liquid simulator, a river mesh generator etc and will be fully integrated and realized in the upcoming versions towards the final release. As with other major assets (**Sky Master ULTIMATE, InfiniGRASS**) this new system is planned as a framework to cover all liquid – river needs, so will be regularly upgraded with new features and refinements long after the initial release. Any feedback and suggestions are also welcome.

For any questions and suggestions please contact me in my **Discord channel** or in the **ARTnGAME Email.**

For quick start on the main system, please refer to my tutorial videos in the following link:
InfiniRIVER Tutorial videos

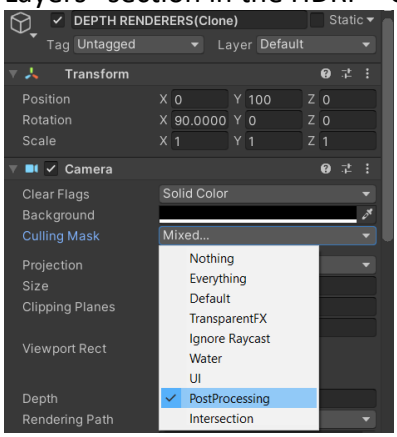
InfiniRIVER Setup steps

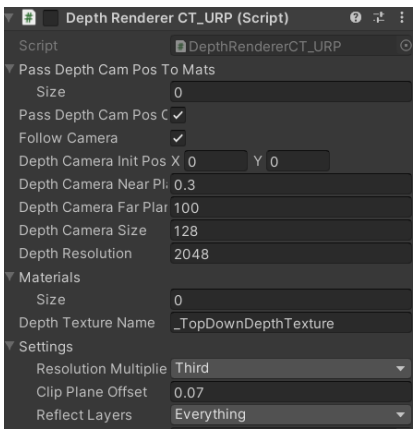
The setup of the system is done by dragging a number of prefabs in the scene. The prefabs contain all required scripts and objects for the full function of the system.

There is also a fully automated configurator module ("InfiniRIVER_Water_SETUP" script) which can instantiated all required prefabs and configured them afterwards for various water sizes and modes (e.g. **follow or not the camera** for endless words, **use world or local space for applying the fluid solver solution** to the water surface). This system is covered after the individual modules details.

Before the setup a layer should be added. **Add a layer to the project** that will be assigned to any objects in the scene that need to interact with the water. By default the 8th Layer slot is assigned to the prefab depth rendering cameras.

If a different layer is needed, it should be manually changed directly in the depth renderer camera in the Standard Pipeline culling mask (left image) or in the depth rendering scripts "Settings", "Reflect Layers" section in the HDRP - URP case.



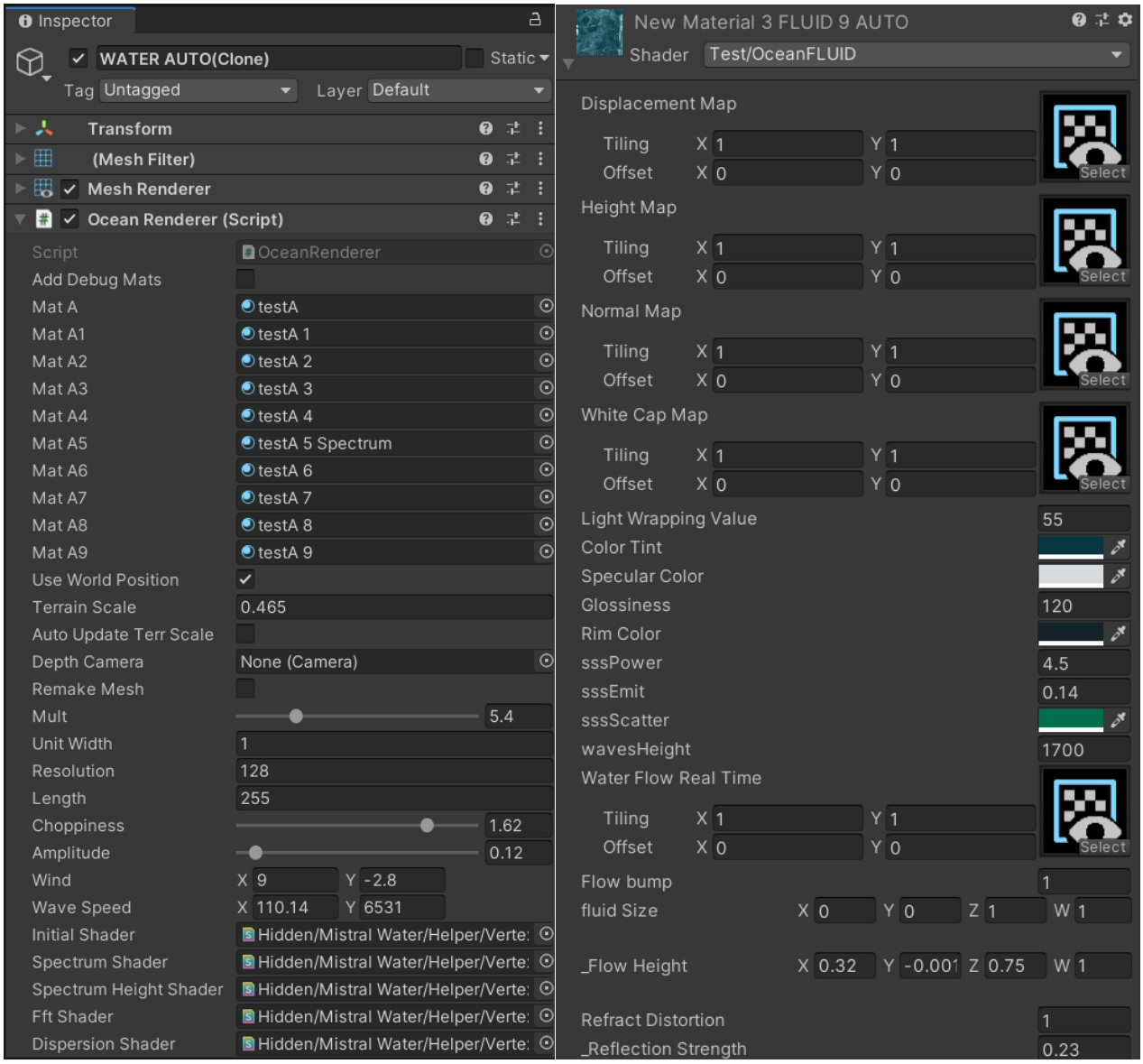


Water mesh creator and Fast Fourier Transform (FFT) based waves generator and controller

The Ocear Renderer module handles the **water mesh generation**. The water is automatically generated based on the Resolution and Length chosen and can be recreated using the “Remake mesh” checkbox.

The module also applies the correct scaling to the water material for the use of fluid solution and enables the world space mode in the water handling. In World Space mode the fluid solution will stay in place when the water surface is moved around the map. In local space the solution follows the water surface. The two modes can be used in various ways to fill a map with various style of rivers and waterfalls.

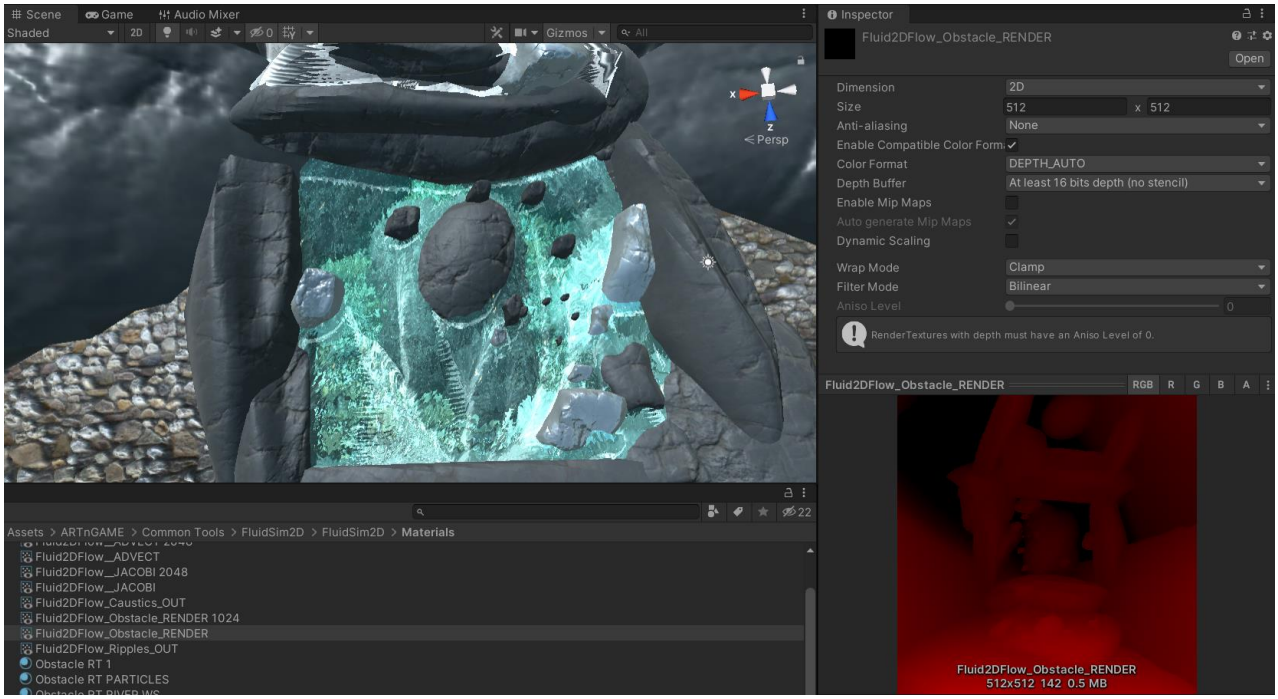
The wave pattern is also controlled by the module, using the “Mult” (Water frequency), “Choppiness” (wave edge amount) and “Amplitude” (wave height) variables. The “Wind” will apply a directional extra wave control based on the wind direction and amount and the “Wave speed” can apply a overall directional motion to the waves.



The water material uses the “OceanFLUID” shader and the water Ocean Renderer module applies the Displacement, Height, Normal and White Cap maps to the material directly. The rest of textures on the material reference the various output textures of the fluid and ripples solvers.

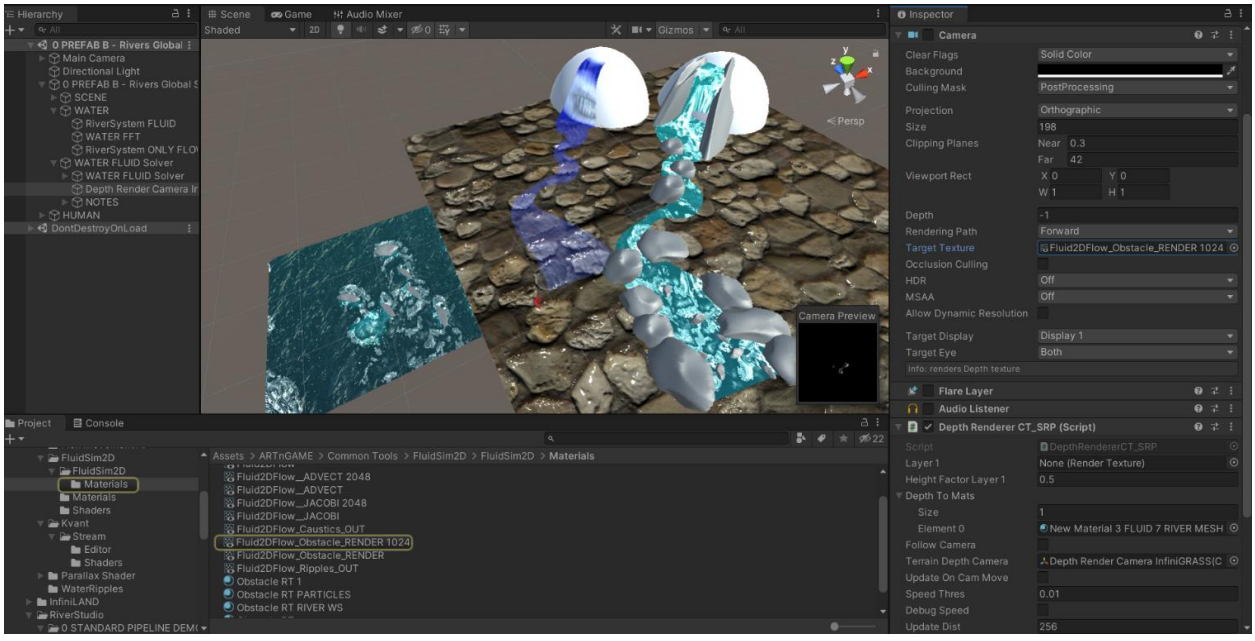
Depth rendering of scene top-down for defining Interactors with water

The system uses a top-down view depth rendering camera to render obstacles that will interact with water fluid simulator. This camera renders the items in the chosen layers to a rendertexture that is then applied to a material used by the fluid simulator to add obstacles to the fluid emulation. Note the obstacles render texture is not directly referenced in the water material, rather the fluid solution that has taken the obstacles into account is used instead to show the complete fluid simulation.

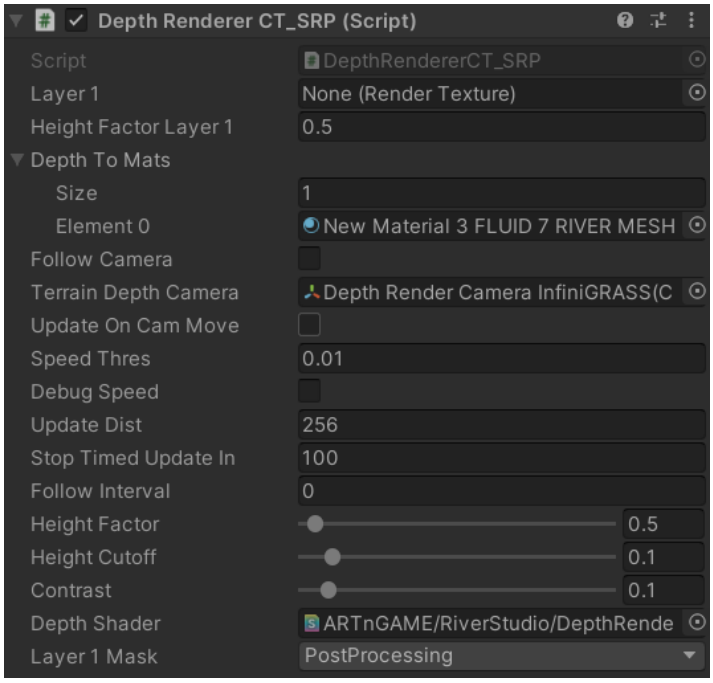


Depth rendering render texture, used to determine the obstacles used by the fluid simulator.

Depth rendering camera and control script (Standard Pipeline):



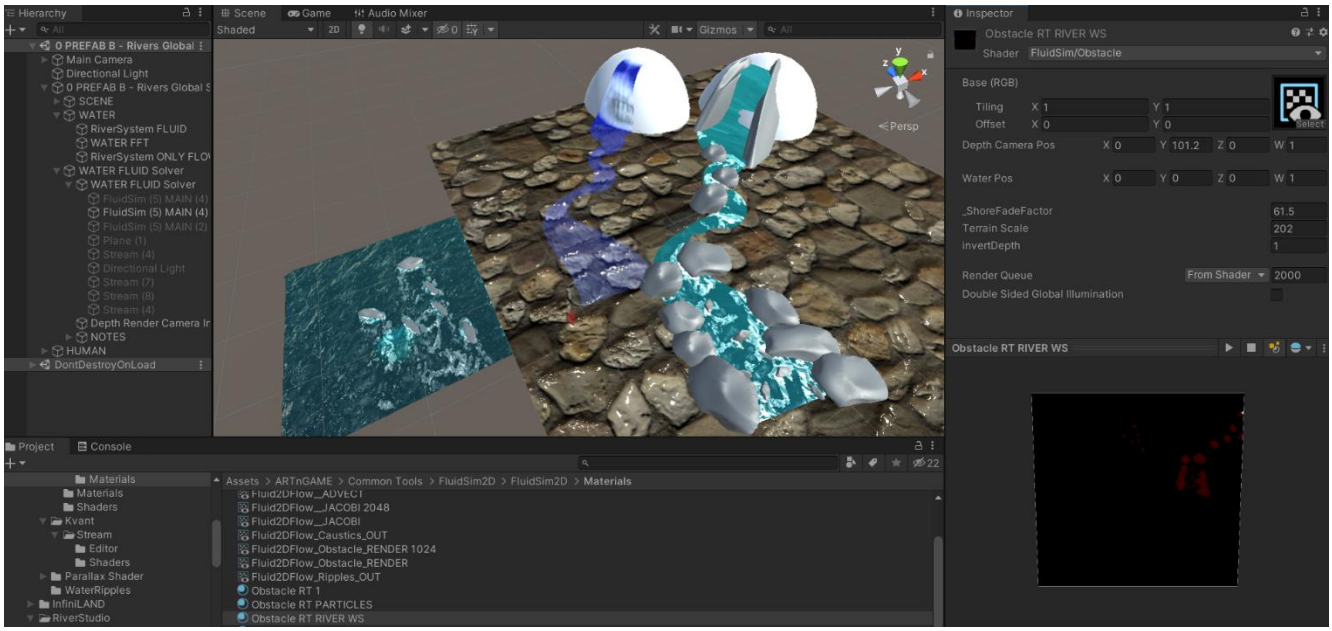
The target depth render texture assigned to the depth rendering camera. This texture is then referenced to the Obstacles material “Base” texture, and this material is then referenced in the fluid simulator in “ObstaclesMat” slot.



The depth rendering system has two modes, the **follow camera mode** will make the depth camera have the same X-Z position as the main scene camera, thus render the water fluid obstacles around the player in a dynamic way for infinite maps. Note that this mode works best with regulated player speed, as the player motion induces an opposite fluid interaction with water and obstacles, due to the fast relative change of obstacles in the fluid solver.

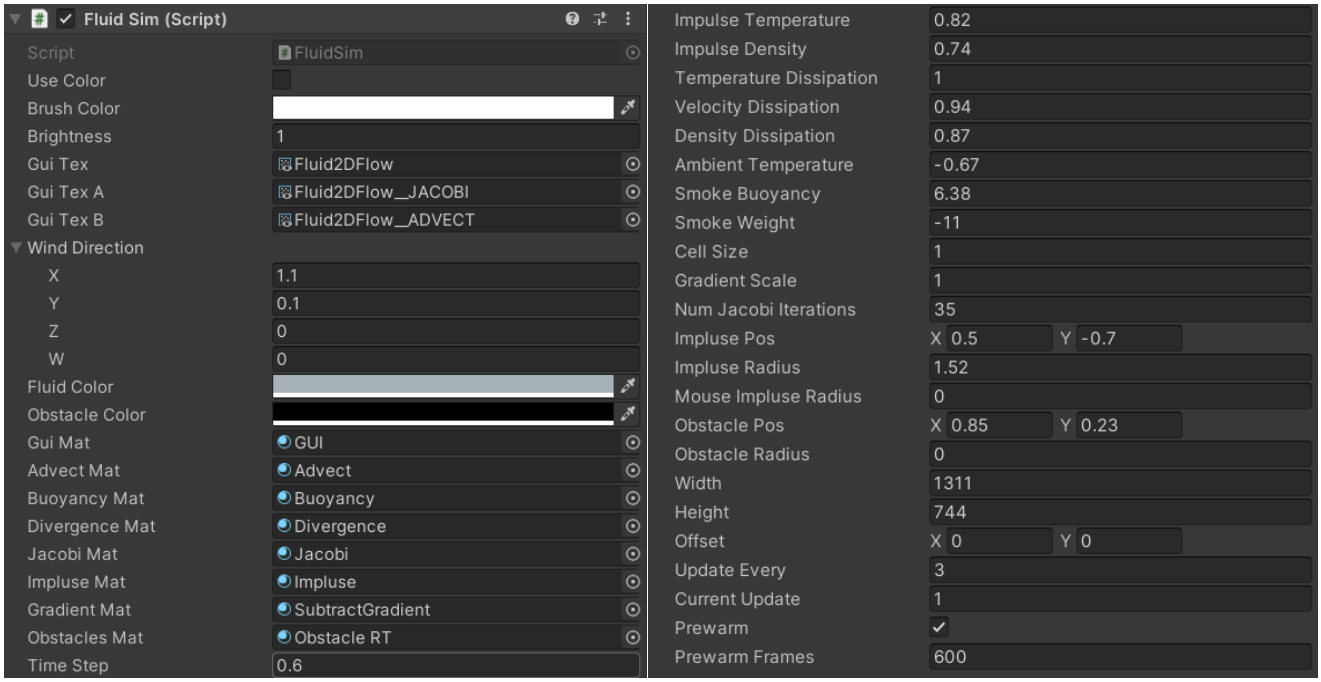
The water material must be referenced in the “DepthToMats” slot so the system can pass the correct depth camera position to the water shader, so it knows where to place the rendered fluid solution. **This is required when the World Space mode is used**, in **Local Space mode** there is no need to reference the material. If follow mode is not activated, the top-down rendering will happen only for the place where the depth camera is initially placed.

For World space obstacle rendering, the Depth camera pos X-Z and Water Pos must be set to zero. The rest of the values are auto-adjusted by the configurator module at setup time and at run time by the ocean creation – FFT wave generator script.

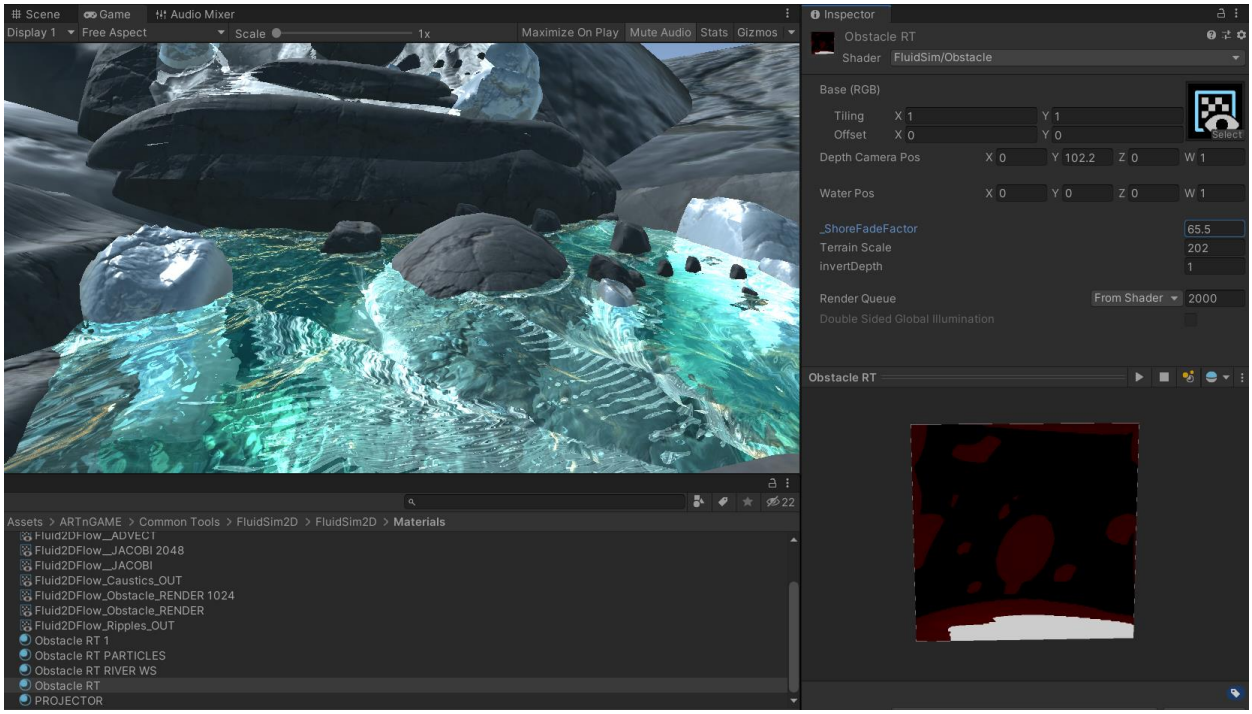


Real time Fluid Simulation

The water fluid simulator is shown in the following image



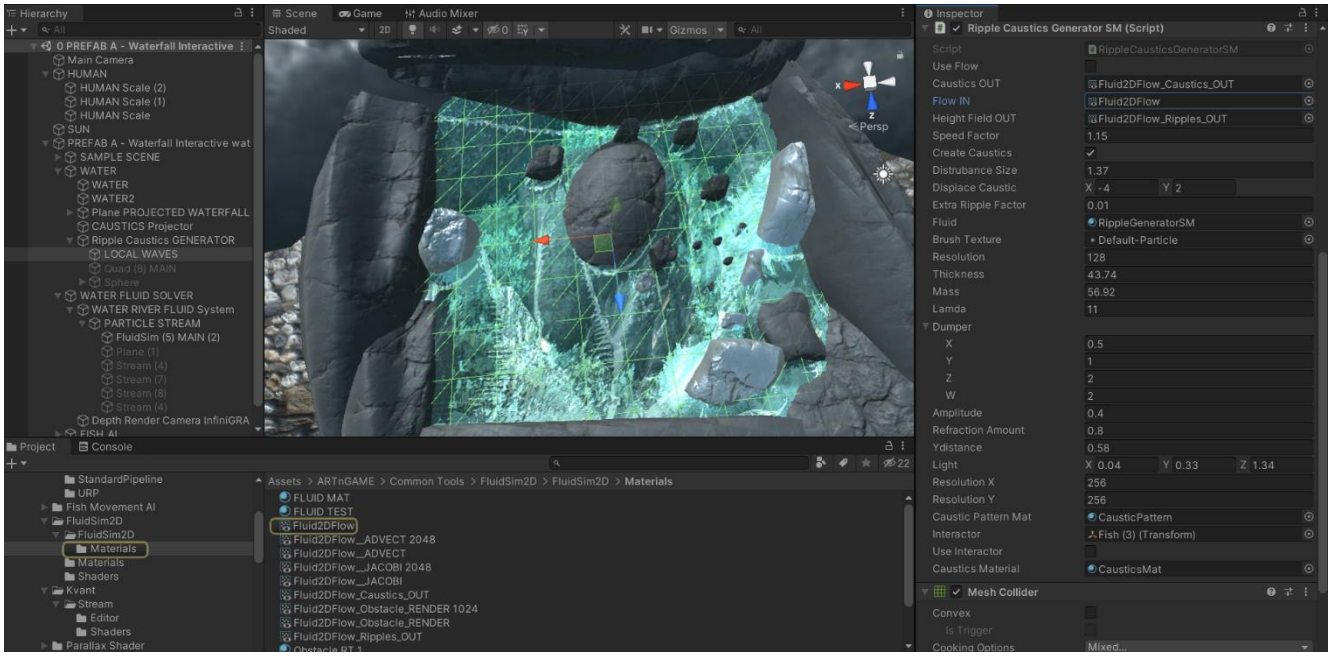
The setup of the system uses a number of materials that handle the various stages of the simulation. These are pre-configured in the Fluid Simulator prefab. The **material that is user adjusted** is the **“Obstacles Mat”,** in this material **the top-down depth rendering camera render texture** is assigned in the **“Base”** texture slot, in order to provide the final obstacles map for the fluid solver, as shown in the image below and mentioned in the depth renderer section above.



The fluid simulation will take into account the obstacles shown in the material preview (right image).

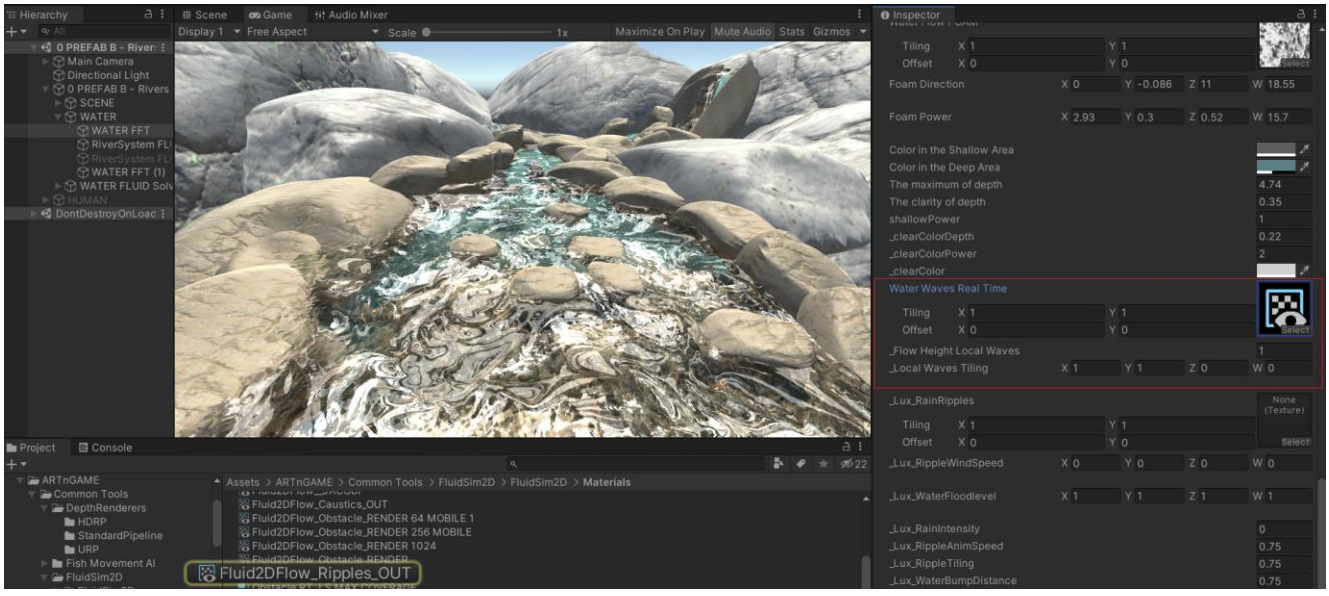
Water ripples and caustics generator

The system includes a module that can generate caustics and water ripples and on the water surface. The input to the system is the interact position by mouse click or drag and the fluid solver render texture output “Fluid2DFlow” in the “Flow IN” slot as shown in the image below. **The Flow input usage is still WIP in the Beta.**



The system outputs two rendertextures, assigned in the “Caustics OUT” and “HeightField OUT” slots, one with the calculated caustics that can be applied to projectors in the scene and another with the water ripples that are applied in the water and ground materials shader.

The “HeightField OUT” rendertexture may be referenced in the water material, in the slot “WaterWaves Real-Time” as shown in the following photo. The “_FlowHeighLocalWaves” can be regulated to adjust the wave height on the water and the tiling is used to adjust the texture placement. **This is auto-adjusted in the automated configurator.**

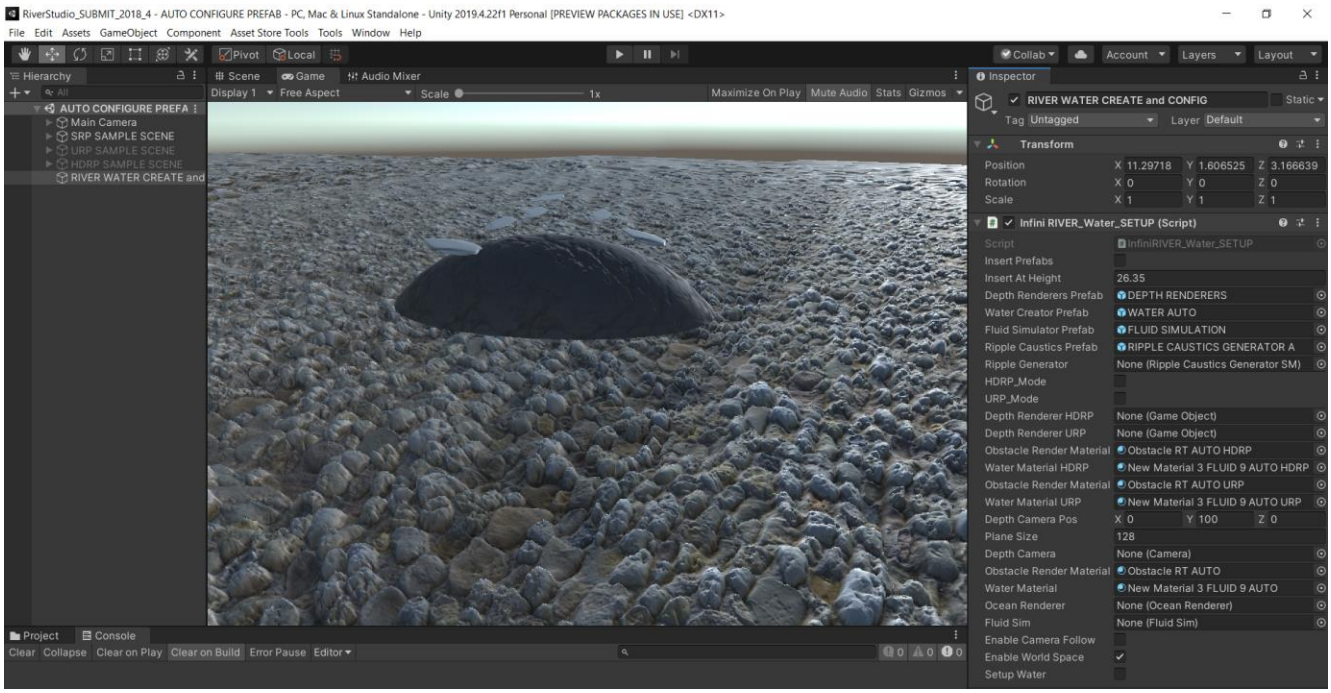


In the coming v1.1 a sample script to control and create the ripples based on scene objects position and speed will be added.

Automated Configuration module

The system includes a configurator **module ("InfiniRIVER_Water_SETUP" script)** which can instantiated all required prefabs and configure them afterwards for various water sizes and modes (e.g. **follow or not the camera** for endless words, **use world or local space for applying the fluid solver solution** to the water surface).

Also a scene has been added with the script ready for use in the various pipelines.

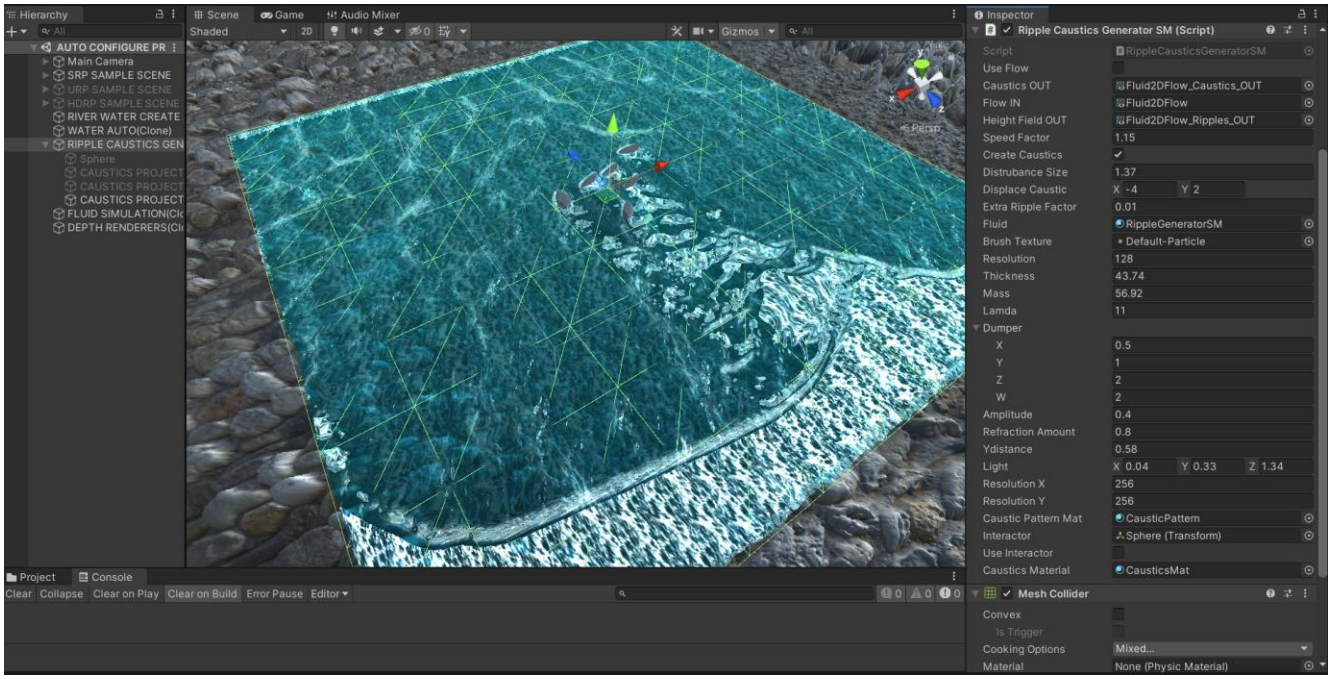


First open the demo scene or insert the "InfiniRIVER_Water_SETUP" script in an empty gameobject.

Then press the "Insert prefabs" checkbox to instantiate the water related prefabs in the scene, as shown below. The instantiated items are also referenced in the configuration script, so can be reconfigured using the various script options.



The system will add the water generator and controller, the fluid simulator, the depth renderers and the ripples and caustics generator modules. Below is shown the general idea of the dynamic ripples – caustics module, a planar collider is used to interact using the mouse with the water and pass the mouse click or drag locations to the system. The output textures are directly applied to a caustics projector, sample projectors are included for each of the pipelines.



After the initial instantiation of the system prefabs, the system can be reconfigured for the various use modules (follow or not the player), water plane sizes and fluid solution application modes (world or local space).

Use the "Plane size" to adjust the water plane size, this value must **correspond to the actual size of the water plane in the scene,** after any extra manual scaling of the plane generated by the Ocean Renderer module.

Use "**Enable Camera follow**" to force the depth rendering cameras follow the main scene camera, such that can cover an endless world, if needed.

Use "**Enable World Space**" checkbox to enable the mode where the fluid solution will stay static even when the water mesh that is applied on moves around or is anywhere in the scene. Id not checked the Local Space mode is applied, where the fluid solution will follow the water mesh it is applied on.

Use the "HDRP_**Mode**" or "URP_**Mode**" to enable either pipeline, if none is checked the system will be setup for the Standard Pipeline. Make sure to first enable the pipeline in the Graphics and Quality Unity settings.

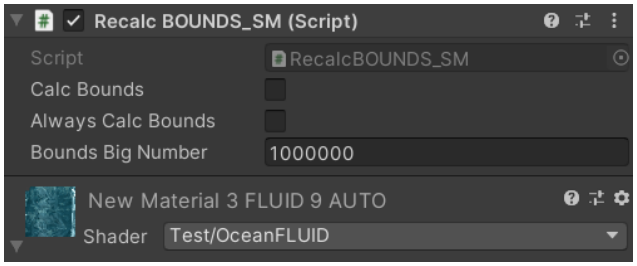
After selecting and adjusting the above options, enable the **"Setup Water"** checkbox, to setup the water with the selected features.

Bonus features

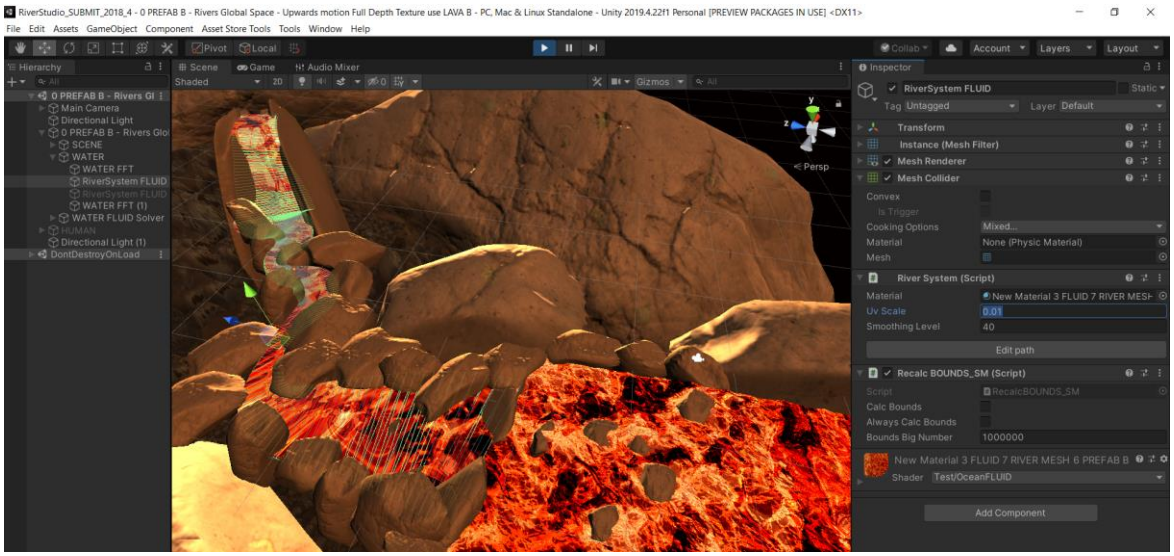
The system contains various bonus features, like a Parallax shader - textures, a river mesh generator and a rock generator. These modules will be fully realized and adapted in the upcoming Beta versions and currently can be used to create some interesting rocky backgrounds for use with the rivers.

Mesh bounds re-calculator

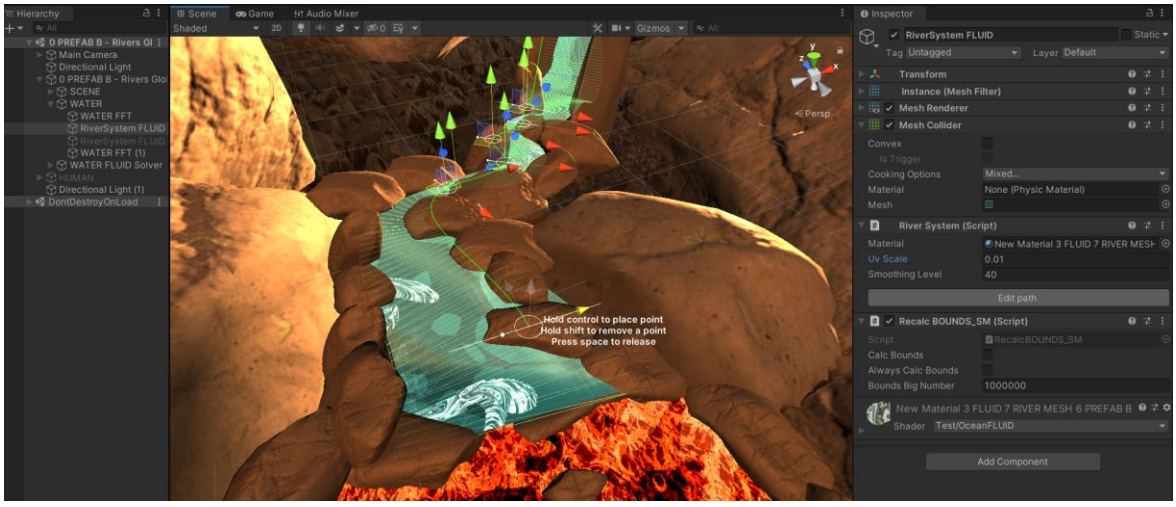
The water meshes generated by the Ocean Renderer may have variable bounds due to the shader vertex manipulation and in some instances may affect the refraction texture. Place the **"Recalc BOUNDS_SM**" script in the water mesh object and select the "**Calc Bounds**" option to assign the bounds to the mesh when the game starts.



River mesh generator.

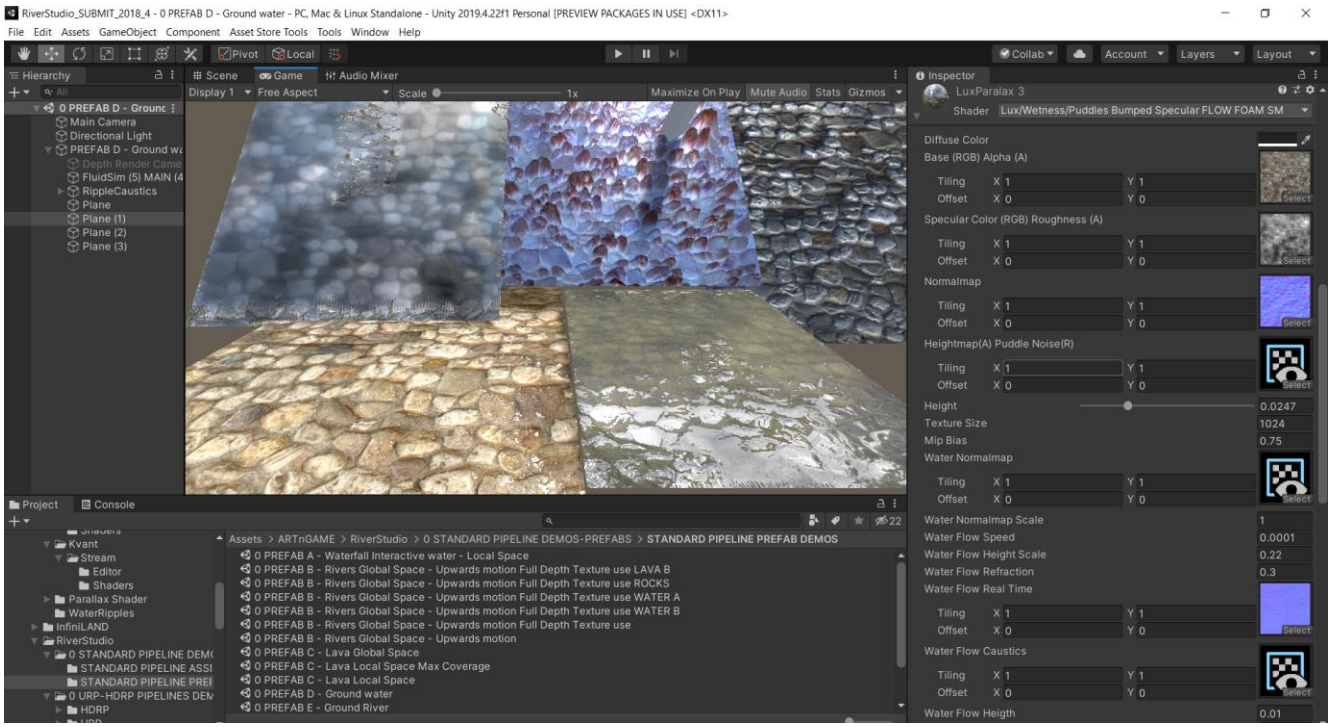


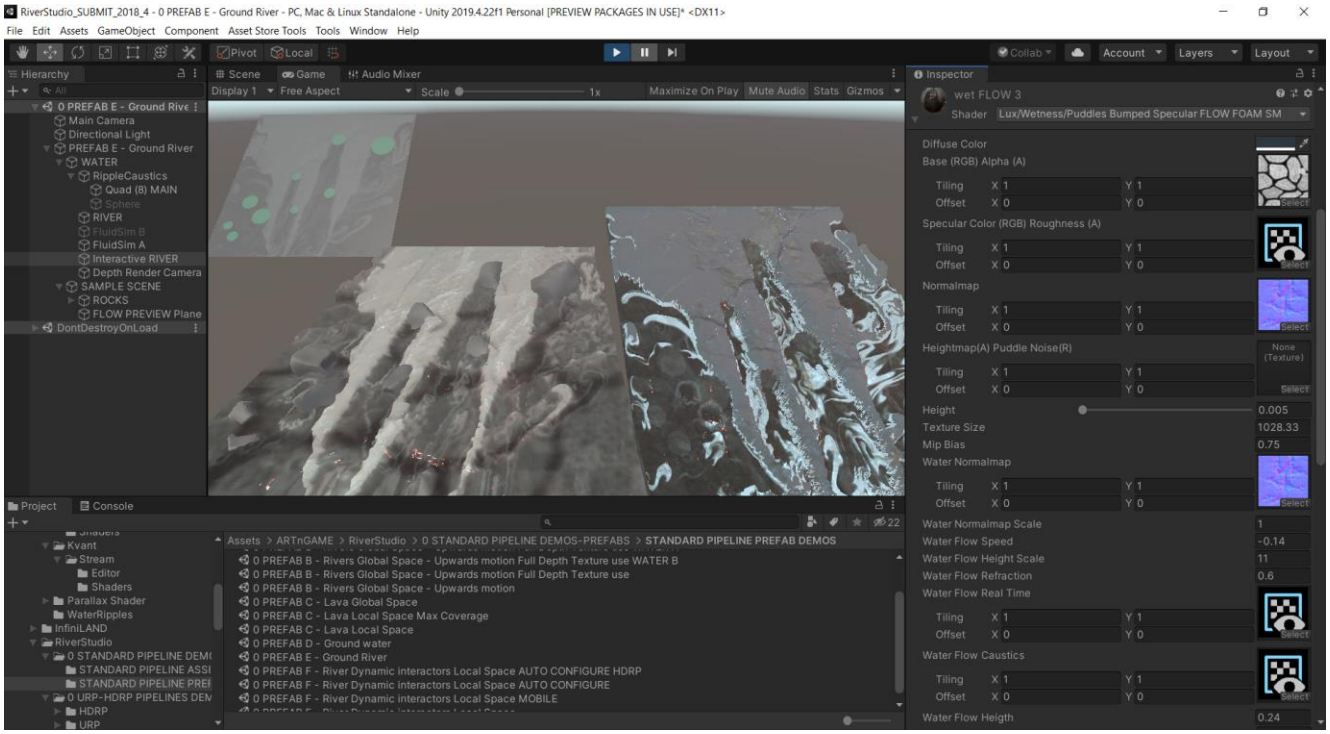
Insert the **"River System" script** to a gameobject and then **press "Edit Path"** to edit the river mesh. Use higher smoothing Level to make the mesh more detailed.



Ground shaders with fluid simulation support

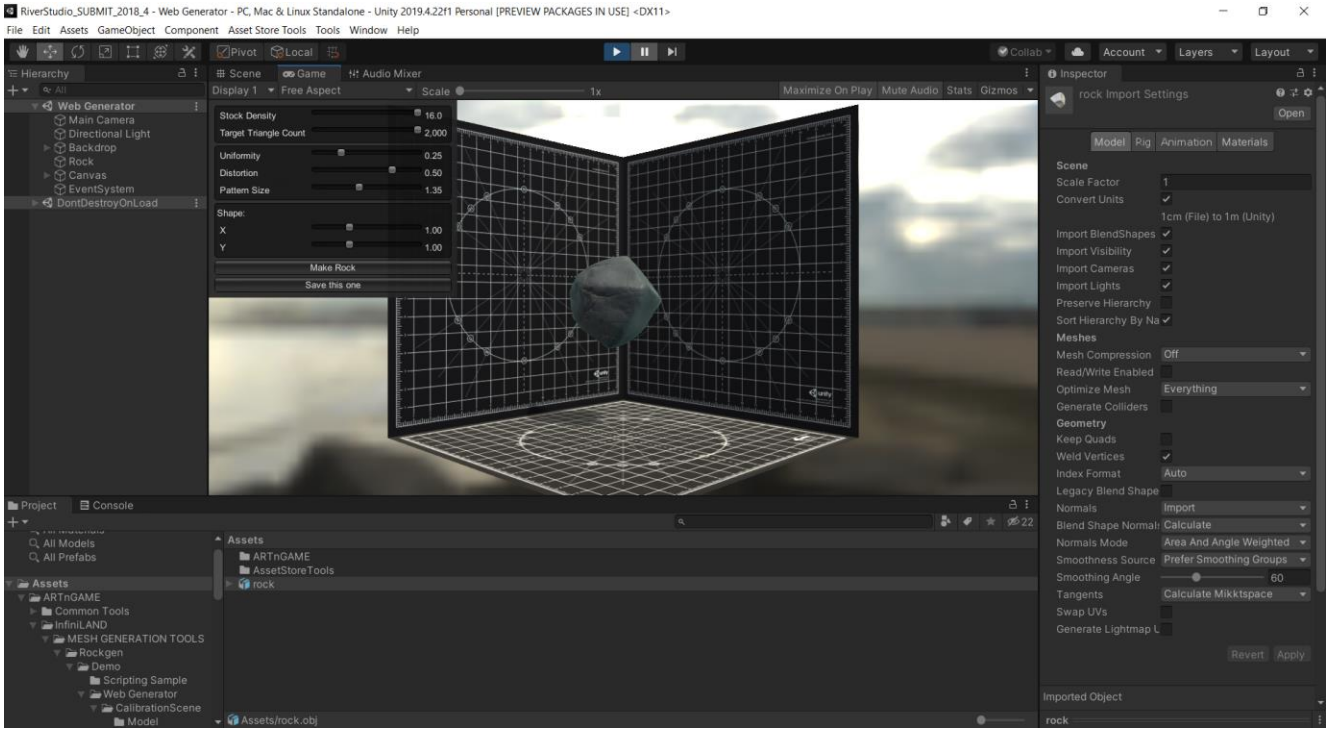
A number of **shaders that can use the fluid simulator output render textures** and the **ripple – caustics generation system outputs** for extra effects are **included for the Standard Pipeline**, those can be found in the demo scenes shown in the following photos. These materials embed the water and water ground in a single shader, for example when shallow water is required.





Rock Generator

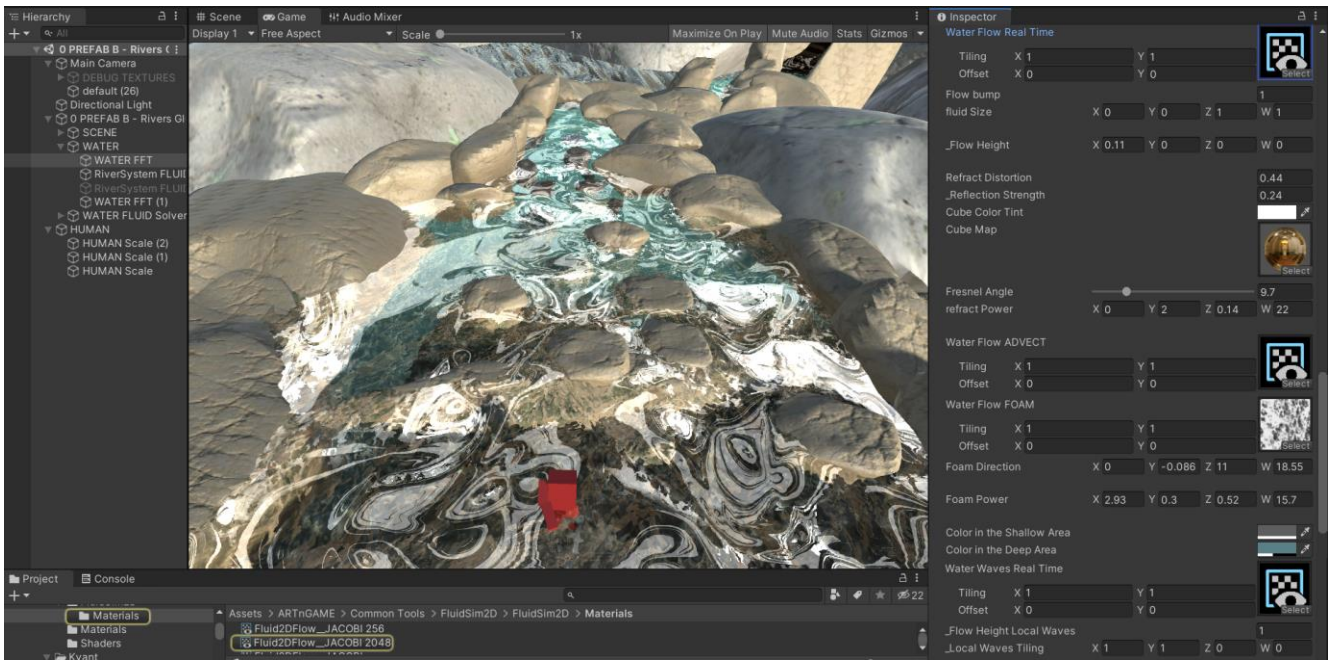
The system includes a rock generator, the module can be accessed in the demo scene shown in the following photo, in the Rockgen – Demo – Web Generator folder.



Enter play mode and adjust the various settings to get unique rocks **and then press "Save this one"** to save the rock in the Assets root folder as shown above with the saved "rock.obj" object.

Water Shader Reference

In this section is presented the water shader and the various options related to adapting the outputs of the fluid simulation and ripples generator to the water.

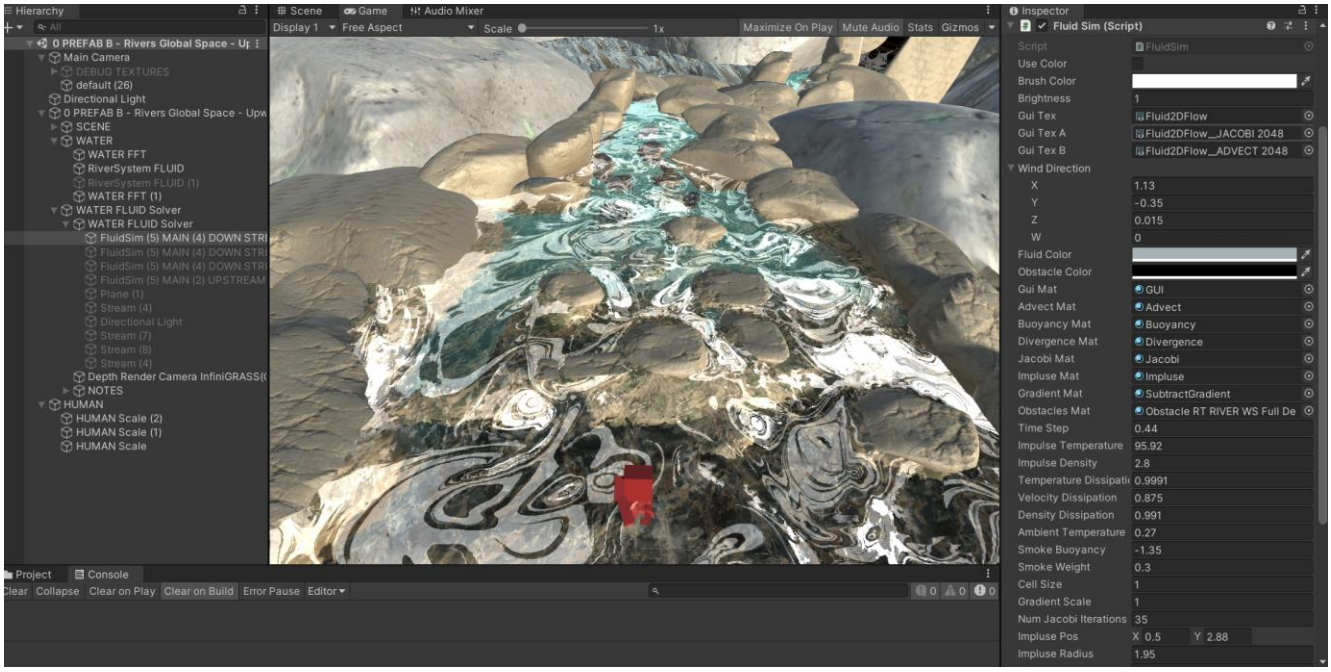


The “**Fluid2D_JACOBI**” render texture from the “**FluidSim**” script should be referenced in the “**Water Flow Real Time**” water material slot.

The “**Fluid2D_FLOW_ADVECT**” render texture from the FluidSim should be relferenced in the “**Water Flow ADVECT**” slot.

The “**Fluid2D_FLOW_Ripples_OUT**” render texture from the ripples and caustics generator script should be referenced in the “**Water Waves Real Time**” slot.

These render texture will pass the result of the fluid simulator and ripples generator to the water.



Known issues:

Issue 1: The URP and HDRP demo scene have pink materials.

Solution 1: Setup either the HDRP or URP pipeline in the Graphics and Quality settings in the Unity settings. Note this will render pink the materials of the other two pipelines, which is the normal intended behavior of Unity in the multiple pipeline use case.

Roadmap (Some of the features are TBD and under evaluation):

- Adapt HDRP Decal shader to have better control of the projected caustic texture
- Control and create water ripples - caustics based on scene objects position – speed, besides mouse motion
- Convert bonus parallax sample shader and parallax with fluid shader to URP-HDRP (**TBD**)
- Add Delauney based tessellation to the river mesh generator
- Add river branching and Terrain river curving
- Add more options in the river mesh maker
- Make easier to adapt the fluid simulation water flow direction
- Add underwater rendering (**TBD**)
- Add slope based component for automatic waterfall creation
- Add flow component using flow maps, finalize the flow map creator module & adapt to water shader
- Add real time reflections to water shader